Single-Phase Transformer



Basic Electrical and Electronics Engineering

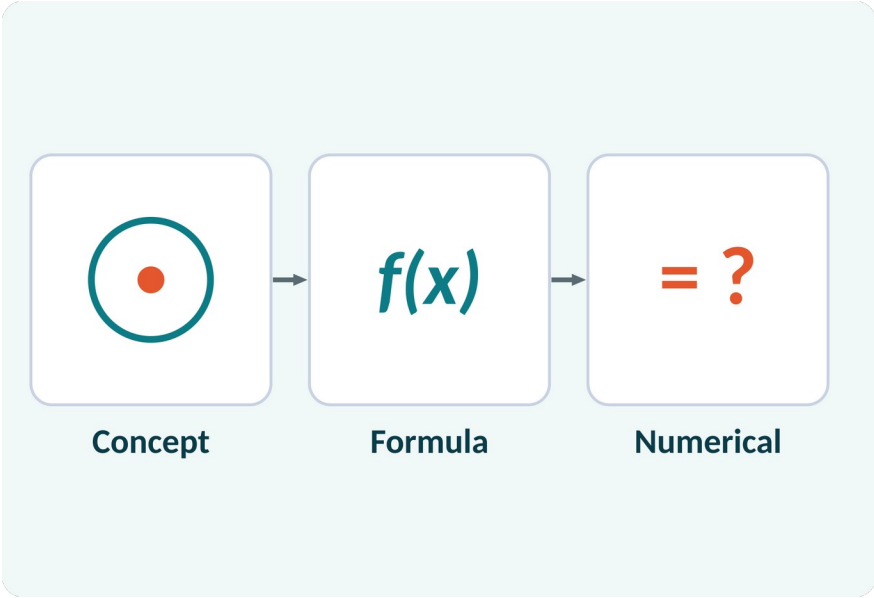
Unit 2: Transformers and Electrical Machines

Topic: Necessity, principle, types, construction, EMF equation, losses and efficiency

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Learning Outcomes

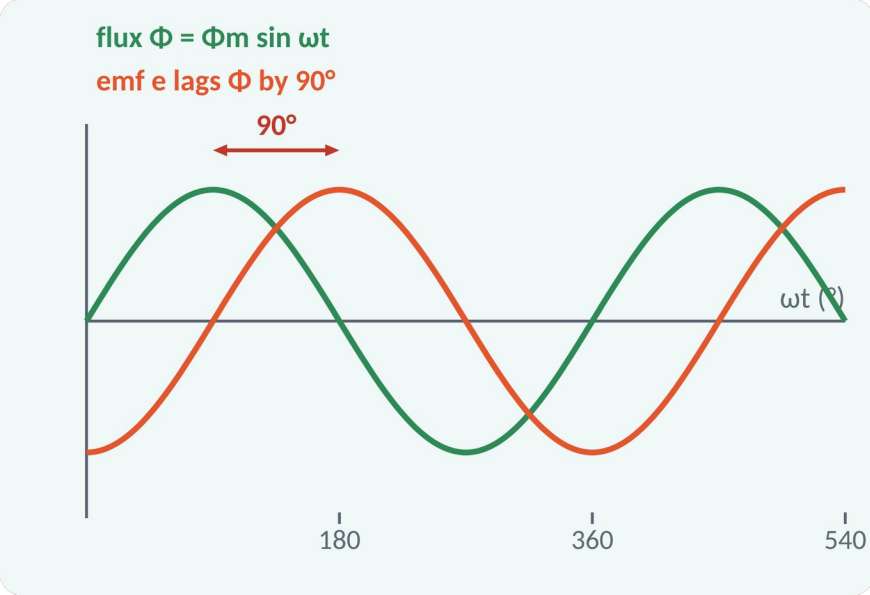
- Explain why transformers are needed.
- Describe principle, construction and types.
- Use the EMF equation and turns ratio.
- Calculate losses and efficiency.



Linked to CO2
Describe the construction and operating principle of transformers.

Recap: What You Already Know

- Faraday: e = −N·dΦ/dt.
- Changing flux induces EMF in a coil.
- Sinusoidal flux gives sinusoidal EMF.
- Mutual induction: flux linking two coils.



e = −N·dΦ/dt; RMS = peak/√2

Motivation: The Phone Charger to the Power Grid

- Charger: 230 V AC → 5 V.
- Generators produce about 11 kV.
- Transmission at 132 kV and above.
- Distribution at 11 kV, then 415/230 V.



Same power P = V·I

10× higher voltage → 10× lower current

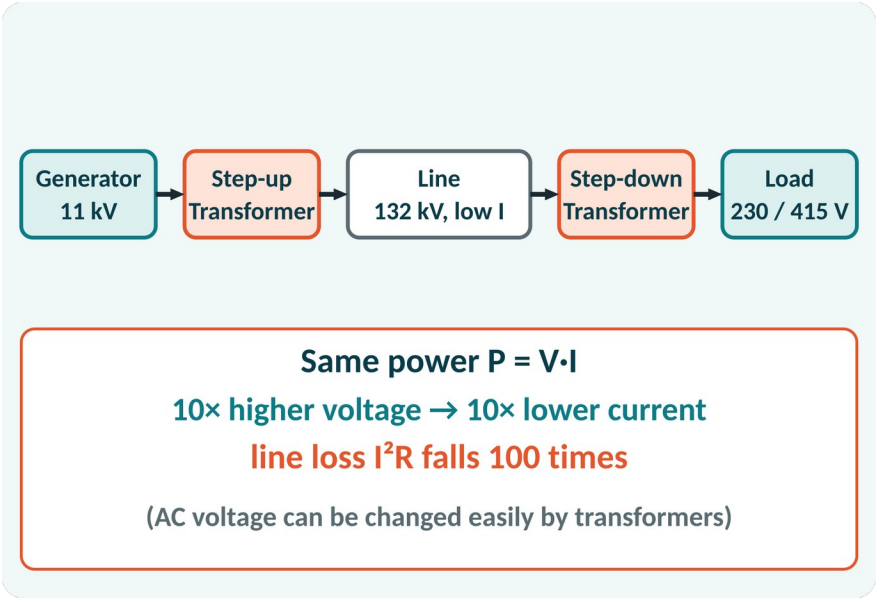
line loss I^2R falls 100 times

(AC voltage can be changed easily by transformers)

Transformers connect all the voltage levels of the grid

Necessity of Transformers

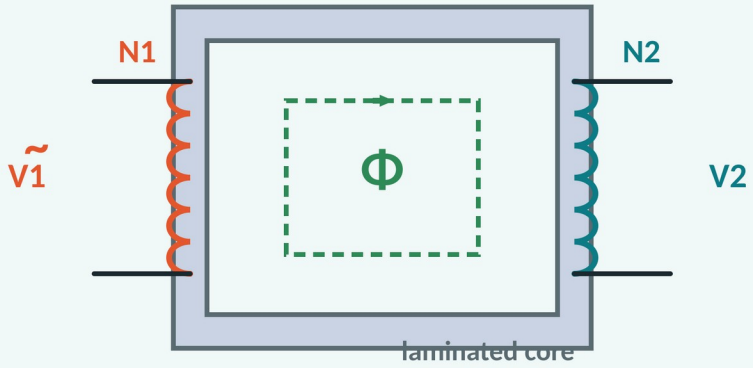
- Power P = V·I stays the same.
- Higher voltage means lower current.
- Line loss is I^2R: it falls sharply.
- Smaller conductors, higher efficiency.
- Safe voltage at the consumer end.



$P_{loss} = I^2R = (P/V)^2R$ → 10× voltage gives 1/100 loss

Principle of Operation

- Primary winding connected to AC supply.
- Alternating flux Φ forms in the core.
- Flux links the secondary winding.
- Mutual induction produces EMF E2.
- Frequency stays the same; no moving parts.

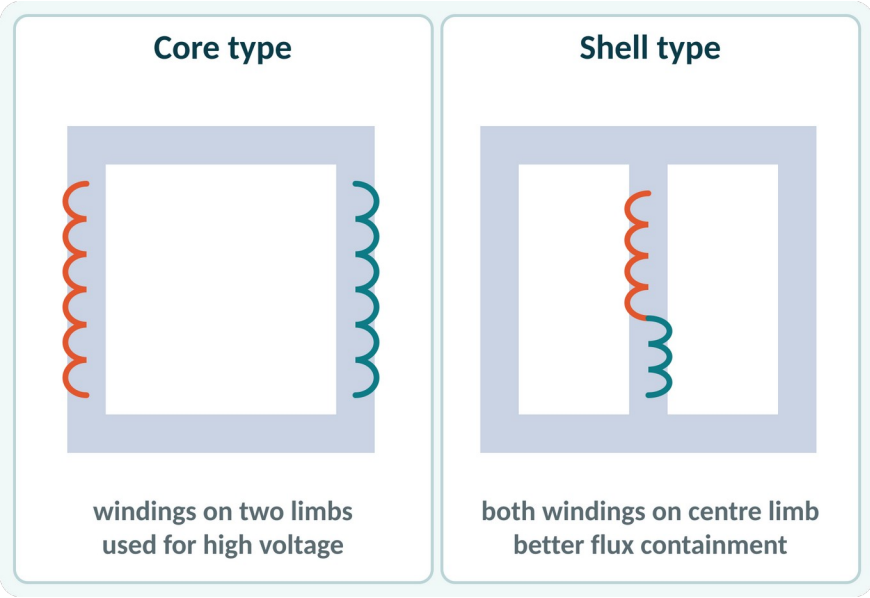


primary: V1 → flux Φ → secondary: E2 (mutual induction)

$$E1 = -N1 \cdot d\Phi/dt, \quad E2 = -N2 \cdot d\Phi/dt \rightarrow E2/E1 = N2/N1$$

Types and Construction

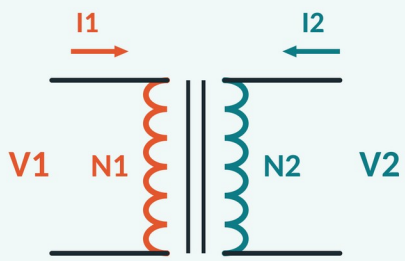
- Core type: windings surround the core limbs.
- Shell type: core surrounds the windings.
- Core of thin silicon-steel laminations.
- Windings: insulated copper or aluminium.
- Oil or air cooling.



Laminations (0.3–0.5 mm) cut eddy-current loss; silicon steel cuts hysteresis loss

Ideal Transformer and Turns Ratio

- No winding resistance, no leakage.
- Core has zero loss and infinite permeability.
- Voltage ratio equals turns ratio.
- Current ratio is the inverse.
- Input power equals output power.



$$V2/V1 = N2/N1 = I1/I2 = K$$

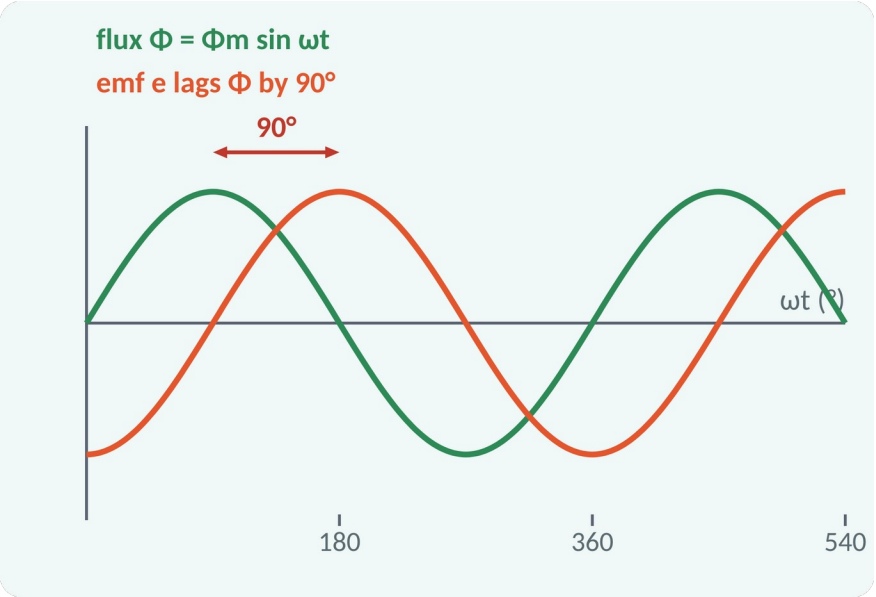
K > 1 step-up; K < 1 step-down

V1·I1 = V2·I2 (ideal: no loss)

$V2/V1 = N2/N1 = I1/I2 = K$; $V1I1 = V2I2$
K transformation ratio

EMF Equation

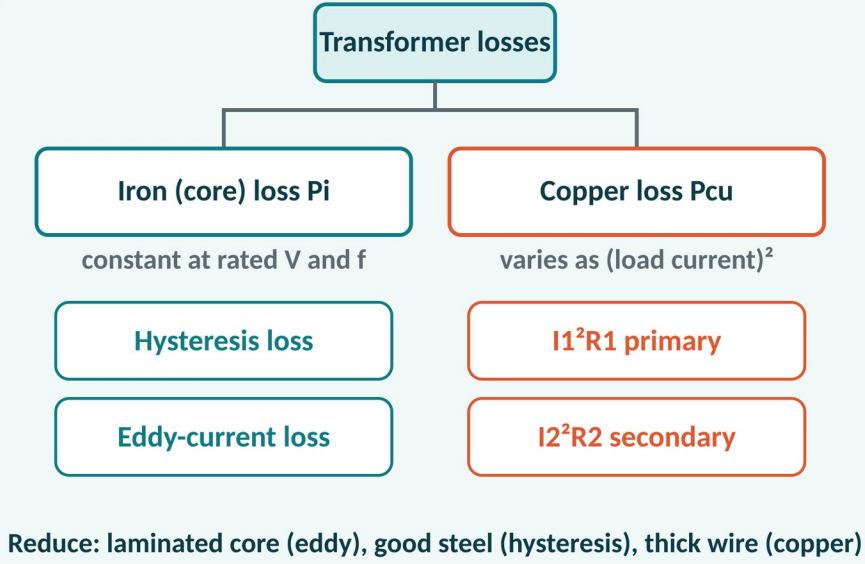
- Flux Φ = Φm sin ωt in the core.
- Induced EMF = N·dΦ/dt, peak ωNΦm.
- RMS value = peak/√2.
- Substitute ω = 2πf.
- Φm = Bm × A.



E = 4.44·f·N·Φm = 4.44·f·N·Bm·A (V)
f Hz, N turns, Φm Wb, Bm T, A m²

Losses in a Transformer

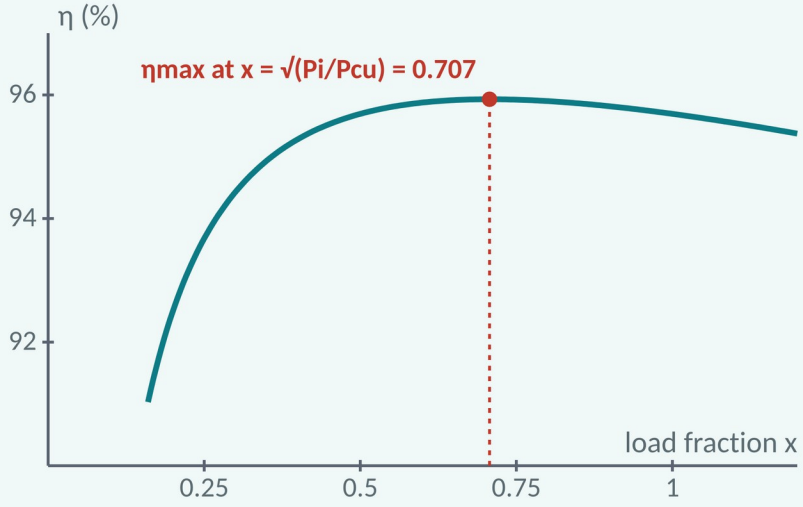
- Iron loss: hysteresis plus eddy current.
- Constant at rated voltage and frequency.
- Copper loss: I^2R in the windings.
- Varies as the square of the load current.
- No rotating parts: no friction loss.



Total loss = $P_i + x^2 \cdot P_{cu,fl}$ (x = load fraction)

Efficiency

- Efficiency = output power / input power.
- Output = x·S·cos φ.
- Maximum efficiency: iron loss = copper loss.
- Load fraction at ηmax: x = √(Pi/Pcu,fl).
- Power transformers: designed for ηmax near full load.

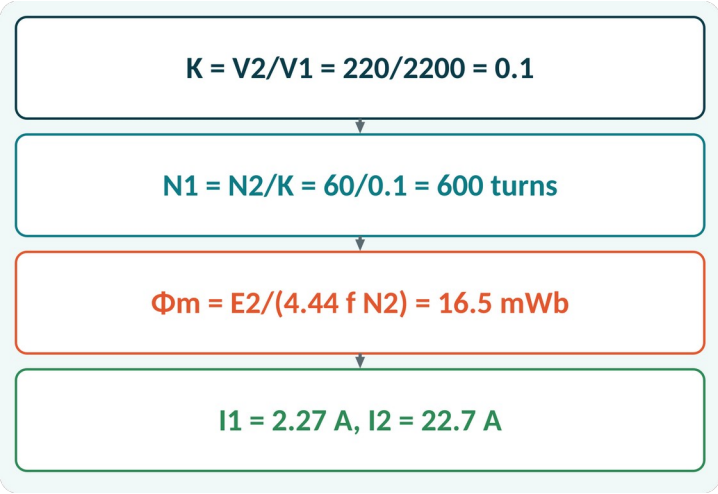


Maximum efficiency when iron loss = copper loss

η = x·S·cos φ / (x·S·cos φ + Pi + x^2·Pcu,fl); ηmax at x = √(Pi/Pcu,fl)

Solved Example 1: Turns and Flux

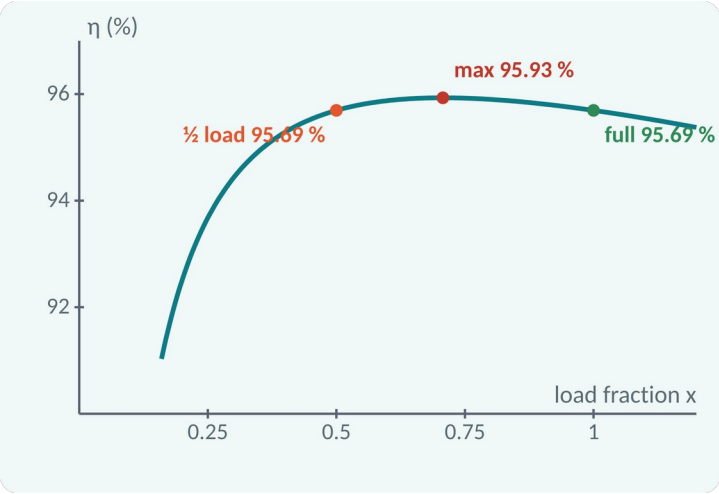
- A 5 kVA, 2200/220 V, 50 Hz transformer has 60 secondary turns. Find N1, Φm, I1 and I2.
- 1. K = 220/2200 = 0.1.
- 2. N1 = N2/K = 600 turns.
- 3. Φm = E2/(4.44·f·N2) = 220/(4.44 × 50 × 60) = 16.5 mWb.
- 4. I1 = 5000/2200 = 2.27 A.
- 5. I2 = 5000/220 = 22.7 A.



Answer: N1 = 600, Φm = 16.5 mWb, I1 = 2.27 A, I2 = 22.7 A

Solved Example 2: Efficiency

- A 25 kVA transformer has iron loss 300 W and full-load copper loss 600 W. For pf 0.8 find η at full load, half load and maximum.
- 1. Full load: output 20 kW; losses 0.9 kW; η = 20/20.9 = 95.69 %.
- 2. Half load: output 10 kW; Pcu = 150 W; η = 10/10.45 = 95.69 %.
- 3. x for ηmax = √(300/600) = 0.707.
- 4. At x = 0.707: output 14.14 kW, losses 0.6 kW, η = 95.93 %.



Answer: 95.69 % (full), 95.69 % (half), ηmax = 95.93 % at 70.7 % load

Common Mistakes and Exam Tips

Common mistakes

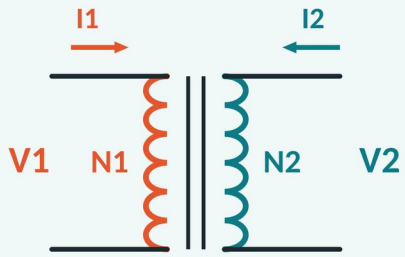
- Using the primary turns for E2 in 4.44fNΦm.
- Mixing kVA with kW.
- Taking copper loss as constant.
- Forgetting x^2 in the copper loss at partial load.
- Applying a DC supply to a transformer.

Exam tips

- Use E = 4.44 f N Φm with the right N.
- Output power = kVA × pf.
- Iron loss: fixed; copper loss: $x^2 \cdot Pcu$.
- Check that V·I is the same on both sides.

Summary

- Transformers change AC voltage with the same frequency.
- Principle: mutual induction in a laminated core.
- V2/V1 = N2/N1 = I1/I2.
- E = 4.44·f·N·Φm.
- ηmax when iron loss = copper loss.



V2/V1 = N2/N1 = I1/I2 = K

K > 1 step-up; K < 1 step-down

V1·I1 = V2·I2 (ideal: no loss)

E = 4.44 f N Φm; η = output/(output + Pi + x^2Pcu)

Quick Quiz

Q1. A transformer works on:
(a) self induction (b) mutual induction (c) static friction (d) DC supply

Ans: (b)

Q2. N1 = 500, N2 = 100, V1 = 230 V gives V2 =
(a) 46 V (b) 1150 V (c) 23 V (d) 100 V

Ans: (a)

Q3. Eddy-current loss is reduced by:
(a) thick core (b) laminations (c) air gap (d) more turns

Ans: (b)

Q4. Maximum efficiency occurs when:
(a) Pi = Pcu (b) Pi = 0 (c) Pcu = 0 (d) no load

Ans: (a)

Q5. Core loss is:
(a) load dependent (b) constant (c) zero (d) $\propto I^2$

Ans: (b)

References

1. **Hughes Edward, Electrical Technology**
   10th Ed., Pearson Education, 2010 (Units 1, 2 textbook)

2. **H. Cotton, Electrical Technology**
   7th Ed., CBS Publishers, 2005 (reference)

Next lecture: DC motor